

1. จงแสดงวิธีทำและหาดีเทอร์มิแนนต์ของ

$$A = \begin{bmatrix} 2 & 0 & 1 & 3 & -2 \\ -2 & 1 & 3 & 2 & -1 \\ 1 & 0 & -1 & 2 & 3 \\ 3 & -1 & 2 & 4 & -3 \\ 1 & 1 & 3 & 2 & 0 \end{bmatrix} \quad \begin{aligned} & 0(C_{11}) + 1(C_{21}) + 0(C_{31}) + (-1)(C_{41}) + 1(C_{51}) \\ & = 0 + 13 + 0 + (-115) + (-33) \\ & = -135 \end{aligned}$$

$$C_{22} = 1 \begin{vmatrix} 2 & 1 & 3 & -2 \\ 1 & -1 & 2 & 3 \\ 3 & 2 & 4 & -3 \\ 1 & 3 & 2 & 0 \end{vmatrix} = 1((-7) + 12 + 8) = 1(13) = 13$$

$$-1 \begin{vmatrix} 1 & 3 & -2 \\ -1 & 2 & 3 \\ 2 & 4 & -3 \end{vmatrix} \begin{vmatrix} 1 & 3 \\ -1 & 2 \\ 2 & 4 \end{vmatrix}$$

$$= (-6) + 18 + 8 + 8 - 12 - 9 = (-1)(7) = -7$$

$$3 \begin{vmatrix} 2 & 3 & -2 \\ 1 & 2 & 3 \\ 3 & 4 & -3 \end{vmatrix} \begin{vmatrix} 2 & 3 \\ 1 & 2 \\ 3 & 4 \end{vmatrix}$$

$$= (-12) + 27 + (-8) + 12 - 24 + 9 = 3(4) = 12$$

$$-2 \begin{vmatrix} 2 & 1 & -2 \\ 1 & -1 & 3 \\ 3 & 2 & -3 \end{vmatrix} \begin{vmatrix} 2 & 1 \\ 1 & -1 \\ 3 & 2 \end{vmatrix}$$

$$= 6 + 9 + (-4) - 6 - 12 + 3 = -2(-4) = 8$$

$$C_{42} = -1 \begin{vmatrix} 2 & 1 & 3 & -2 \\ -2 & 3 & 2 & -1 \\ 1 & -1 & 2 & 3 \\ 1 & 3 & 2 & 0 \end{vmatrix} = (-1)(32 + 129 - 46) = -115$$

$$-1 \begin{vmatrix} 1 & 3 & -2 \\ 3 & 2 & -1 \\ -1 & 2 & 3 \end{vmatrix} \begin{vmatrix} 1 & 3 \\ 3 & 2 \\ -1 & 2 \end{vmatrix}$$

$$= 6 + 3 + (-12) - 4 + 2 - 27 = (-1)(32) = 32$$

$$3 \begin{vmatrix} 2 & 3 & -2 \\ -2 & 2 & -1 \\ 1 & 2 & 3 \end{vmatrix} \begin{vmatrix} 2 & 3 \\ -2 & 2 \\ 1 & 2 \end{vmatrix}$$

$$= 12 + (-3) + 8 + 4 + 4 + 18 = 3(43) = 129$$

$$-2 \begin{vmatrix} 2 & 1 & -2 \\ -2 & 3 & -1 \\ 1 & -1 & 3 \end{vmatrix} \begin{vmatrix} 2 & 1 \\ -2 & 3 \\ 1 & -1 \end{vmatrix}$$

$$= 18 + (-1) + (-4) + 6 - 2 + 6 = (-2)(23) = -46$$

$$C_{52} = -1 \begin{vmatrix} 2 & 1 & 3 & -2 \\ -2 & 3 & 2 & -1 \\ 1 & -1 & 2 & 3 \\ 3 & 2 & 4 & -3 \end{vmatrix} = (-1)((-62) + 12 + (-3) + 86)$$

$$= (-1)(33) = -33$$

$$-1 \begin{vmatrix} -2 & 2 & -1 \\ 1 & 2 & 3 \\ 3 & 4 & -3 \end{vmatrix} \begin{vmatrix} -2 & 2 \\ 1 & 2 \\ 3 & 4 \end{vmatrix}$$

$$= 12 + 18 + (-4) + 6 + 24 + 6$$

$$= (-1)(62) = -62$$

$$3 \begin{vmatrix} 2 & 3 & -2 \\ 1 & 2 & 3 \\ 3 & 4 & -3 \end{vmatrix} \begin{vmatrix} 2 & 3 \\ 1 & 2 \\ 3 & 4 \end{vmatrix}$$

$$= (-12) + 27 + (-8) + 12 - 24 + 9$$

$$= 3(4) = 12$$

$$1 \begin{vmatrix} 2 & 3 & -2 \\ -2 & 2 & -1 \\ 3 & 4 & -3 \end{vmatrix} \begin{vmatrix} 2 & 3 \\ -2 & 2 \\ 3 & 4 \end{vmatrix}$$

$$= (-12) + (-9) + 16 + 12 + 8 - 18$$

$$= 1(-3) = -3$$

$$2 \begin{vmatrix} 2 & 3 & -2 \\ -2 & 2 & -1 \\ 1 & 2 & 3 \end{vmatrix} \begin{vmatrix} 2 & 3 \\ -2 & 2 \\ 1 & 2 \end{vmatrix}$$

$$= 12 + (-3) + 8 + 4 + 4 + 18$$

$$= 2(43) = 86$$

2. จงแสดงวิธีทำและหาค่า $|A^{-1}|$

นั่นคือ เริ่มต้นด้วยการหา A^{-1} จากนั้นให้ทำการหาดีเทอร์มิแนนต์ของเมทริกซ์นั้น

$$A = \begin{bmatrix} 1 & 0 & -1 & 3 \\ 1 & 0 & 3 & -2 \\ 2 & 0 & 2 & -1 \\ 1 & -3 & 1 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det(A)} \times \text{adj}(A)$$

$$\det(A) = 0(C_{12}) + 0(C_{22}) + 0(C_{32}) + (-3)(C_{42})$$

$$C_{42} = \begin{vmatrix} 1 & -1 & 3 \\ 1 & 3 & -2 \\ 2 & 2 & -1 \end{vmatrix} = -8 \quad \det(A) = (-3)(-8) = 24$$

$$\text{adj}(A) = \begin{bmatrix} C_{11} & C_{12} & C_{13} & C_{14} \\ C_{21} & C_{22} & C_{23} & C_{24} \\ C_{31} & C_{32} & C_{33} & C_{34} \\ C_{41} & C_{42} & C_{43} & C_{44} \end{bmatrix}^T$$

$$C_{11} = \begin{vmatrix} 0 & 3 & -2 \\ 0 & 2 & -1 \\ -3 & 1 & 2 \end{vmatrix} = -3$$

$$C_{21} = - \begin{vmatrix} 1 & -1 & 3 \\ 0 & 2 & -1 \\ -3 & 1 & 2 \end{vmatrix} = -15$$

$$C_{12} = \begin{vmatrix} 1 & 3 & -2 \\ 2 & 2 & -1 \\ 1 & 1 & 2 \end{vmatrix} = 10$$

$$C_{22} = \begin{vmatrix} 1 & -1 & 3 \\ 2 & 2 & -1 \\ 1 & 1 & 2 \end{vmatrix} = 10$$

$$C_{13} = \begin{vmatrix} 1 & 0 & -2 \\ 2 & 0 & -1 \\ 1 & -3 & 2 \end{vmatrix} = 9$$

$$C_{23} = \begin{vmatrix} 1 & 0 & 3 \\ 2 & 0 & -1 \\ 1 & -3 & 2 \end{vmatrix} = 21$$

$$C_{14} = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 0 & 2 \\ 1 & -3 & 1 \end{bmatrix} = 12$$

$$C_{24} = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 0 & 2 \\ 1 & -3 & 1 \end{bmatrix} = 12$$

$$C_{31} = \begin{bmatrix} 0 & -1 & 3 \\ 0 & 2 & -2 \\ -3 & 1 & 2 \end{bmatrix} = 21$$

$$C_{32} = -\begin{bmatrix} 1 & -1 & 3 \\ 1 & 3 & -2 \\ 1 & 1 & 2 \end{bmatrix} = -6$$

$$C_{33} = \begin{bmatrix} 1 & 0 & 3 \\ 1 & 0 & -2 \\ 1 & -3 & 2 \end{bmatrix} = -15$$

$$C_{34} = -\begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & 3 \\ 1 & -3 & 1 \end{bmatrix} = -12$$

$$C_{41} = -\begin{bmatrix} 0 & -1 & 3 \\ 0 & 3 & -2 \\ 0 & 2 & -2 \end{bmatrix} = 0$$

$$C_{42} = \begin{bmatrix} 1 & -1 & 3 \\ 1 & 3 & -2 \\ 2 & 2 & -1 \end{bmatrix} = -8$$

$$C_{43} = -\begin{bmatrix} 1 & 0 & 3 \\ 1 & 0 & -2 \\ 2 & 0 & -1 \end{bmatrix} = 0$$

$$C_{44} = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & 3 \\ 2 & 0 & 2 \end{bmatrix} = 0$$

$$A^{-1} = \frac{1}{24} \times \begin{bmatrix} -3 & -15 & 21 & 0 \\ 10 & 10 & -6 & -8 \\ 9 & 21 & -15 & 0 \\ 12 & 12 & -12 & 0 \end{bmatrix} = \begin{bmatrix} \frac{-1}{8} & \frac{-5}{8} & \frac{7}{8} & \mathbf{0} \\ \frac{5}{12} & \frac{5}{12} & \frac{-1}{4} & \mathbf{\frac{-1}{3}} \\ \frac{3}{8} & \frac{7}{8} & \frac{-5}{8} & \mathbf{0} \\ \frac{1}{2} & \frac{1}{2} & \frac{-1}{2} & \mathbf{0} \end{bmatrix}$$

$$\det(A^{-1}) = (0)(C_{14}) + (\frac{-1}{3})(C_{24}) + (0)(C_{34}) + (0)(C_{44})$$

$$= \left(\frac{-1}{3}\right) \begin{bmatrix} \frac{-1}{8} & \frac{-5}{8} & \frac{7}{8} \\ \frac{3}{8} & \frac{7}{8} & \frac{-5}{8} \\ \frac{1}{2} & \frac{2}{2} & \frac{-1}{2} \end{bmatrix} = \frac{1}{24}$$